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B.C.A. (Fifth Semester)
Examination, 2026

Second Paper
(JAVA Programming and Dynamic
Webpage Design)

Time : 2:00 Hours] [Maximum Marks : 75

Note : Attempt **all** sections as per instructions.

Section-A

(Very Short Answer Type Questions)

Note : Attempt all the **05 (five)** questions. Each question carries **02 (two)** marks and answer of each question should not exceed **50** words. $5 \times 2 = 10$

1. (a) List two key features of Java's exception handling mechanism.
- (b) Mention any one advantage of multi threading.

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- (c) Name the main method for reading data from a Datagram Socket.
- (d) Give the syntax of an HTML comment.
- (e) Name one standard tag library element used in JSP.

Section-B

(Short Answer Type Questions)

Note : Attempt any **05 (five)** questions out of total **08 (eight)** questions. Each question carries **05 (five)** marks and answer of each question should not exceed **100** words. $5 \times 5 = 25$

2. (a) Explain how inheritance is implemented in Java and mention its types.
- (b) Describe the significance of access specifiers in Java.

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- (c) Discuss how arrays are different from strings in Java.
- (d) Compare datagram sockets and TCP/IP sockets in Java networking.
- (e) What issues can arise with connection pooling in database connections?
- (f) Describe the lifecycle stages of an HTTP servlet. <https://www.mgkvponline.com>
- (g) How does JSP support custom tag libraries with examples?
- (h) Explain the structure and organization of a JSP file.

Section-C

(Long Answer Type Questions)

Note : Attempt any **02 (two)** questions out of total **04 (four)** questions. Each question carries **20 (twenty)** marks and answer of each question should not exceed **400** words. $20 \times 2 = 40$

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3. (a) Implement a Java program using multithreading to compute Fibonacci numbers, demonstrating synchronized blocks.
- (b) Create an HTML page with advanced formatting, including forms, tables, and image maps. Explain your code.
- (c) Illustrate in detail how Java handles event-driven programming for networked applications.
- (d) Design and code Java applet that takes user input for registration and validates entries using listeners.

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